How-To Library

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Importing and Exporting Assets Between Teradata and Enterprise Data Preparation Using the Teradata Connector for Hadoop

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Abstract

You can use the Teradata Connector for Hadoop (TDCH) to import data from a Teradata database into a data lake that is a Cloudera CDH cluster or a Hortonworks HDP cluster. You can also use TDCH to export data from a data lake that is a Cloudera CDH cluster or a Hortonworks HDP cluster to a Teradata database. Review this article to prepare Enterprise Data Preparation to import and export data assets.

Supported Versions

- Enterprise Data Lake 10.2 x
- Enterprise Data Lake 10.4.0

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Overview

You can use the Teradata Connector for Hadoop (TDCH) to import data from a Teradata database into a data lake that is a Cloudera CDH cluster or a Hortonworks HDP cluster. You can also use TDCH to export data from a data lake that is a Cloudera CDH cluster or a Hortonworks HDP cluster to a Teradata database.

After you import a data asset from Teradata, you can add the imported data as a worksheet to a project in Enterprise Data Preparation. After you prepare an asset and export the data to Teradata, you can perform further operations on the data in Teradata.

To prepare to import and export data assets between Teradata and Enterprise Data Preparation, complete the following steps:

1. Download either the Cloudera Connector Powered By Teradata or the Hortonworks Connector for Teradata, and then copy the connector files to the Data Integration Service machine associated with Enterprise Data Preparation. The connector you download depends on whether the data lake is a Cloudera CDH cluster or a Hortonworks HDP cluster.
2. Download the Teradata JDBC Driver, and then copy the driver files to the Data Integration Service machine associated with Enterprise Data Preparation.
3. Create one or more JDBC connections to import and export data assets. You might create different connections for import and export operations.
4. Add the DoNotUseOwnerNameForSqoop custom property to the Enterprise Data Preparation Service.

Download the Connector

Download either the Cloudera Connector Powered By Teradata or the Hortonworks Connector for Teradata, and then copy the connector files to the Data Integration Service machine associated with Enterprise Data Preparation.

The connector you download depends on whether the data lake is a Cloudera CDH cluster or a Hortonworks HDP cluster. The connector enables data transfer between the data lake and a Teradata database.

Extract all of the files from the connector archive, and then copy the files to the following directory on the Data Integration Service machine:

```
<Informatica installation directory>/externaljdbcjars
```

Download the Cloudera Connector Powered By Teradata

Download the Cloudera Connector Powered By Teradata from the following link:

https://cloudera.com

Download the Hortonworks Connector for Teradata

Download the Hortonworks Connector for Teradata from the following link:

https://hortonworks.com/downloads/#data-platform

Download the JDBC Driver

Download the Teradata JDBC driver, and then copy the driver files to the Data Integration Service machine associated with Enterprise Data Preparation.

When you import or export data, the Data Integration Service uses Sqoop to connect to the cluster, and then it runs a Sqoop job on the cluster to import or export the data. Sqoop uses the driver to connect to the Teradata database target when the import or export job runs on the cluster.

Download the JDBC driver archive file from the following link:

http://downloads.teradata.com/download/connectivity/jdbc-driver

Extract the terajdbc4.jar and tdgssconfig.jar files from the archive, and then copy the files to the following directory on the Data Integration Service machine:

```
<Informatica installation directory>/externaljdbcjars
```

Create JDBC Connections to Access a Teradata Database

Create one or more JDBC connections to import and export data assets. You might create different connections for import and export operations.

You include the arguments Sqoop uses to connect to the Teradata database in each JDBC connection. The string you include depends on whether the data lake is a Cloudera CDH cluster or a Hortonworks HDP cluster.

1. In the Administrator tool, click the **Manage** tab > **Connections** view.
2. In the Domain Navigator, click **Actions > New > Connection**.

   The New Connection dialog box appears.
3. In the New Connection dialog box, select **Database** > **JDBC**.
4. Click **OK**.

The **New Connection** wizard appears.

5. Enter the JDBC connection properties required to connect to the Teradata database.

   The following table describes the connection properties you enter:

Property	Description
Name	Name of the connection. The name is not case sensitive, and must be unique within the domain. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: `~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /`
ID	String that the Informatica services use to identify the connection. The string is not case sensitive. The string must be 255 characters or less, and must be unique in the domain. Default value is the connection name.
Description	Description of the connection. The description cannot exceed 765 characters.
User Name	Database user name.
Password	Password for the database user name.
JDBC Driver Class Name	Name of the Teradata JDBC driver class. Enter the following value: `com.teradata.jdbc.TeraDriver`

Connection String	Connection string to connect to the database. Use the following format for the connection string:
	`jdbc:teradata://<database host name>:<port>;SID=<database name>`
	The following example shows a sample connection string:
	`jdbc:teradata://TDVME15.company.com/database=db_01`

Environment SQL	Optional. SQL commands to set the database environment when you connect to the target. The Data Integration Service executes the connection environment SQL each time it connects to the database.
Transaction SQL	Optional. SQL commands to set the database environment when you connect to the target. The Data Integration Service executes the SQL at the beginning of each transaction.
SQL Identifier Character	Select " " from the drop down list to connect to the target from a Cloudera CDH cluster or from a Hortonworks HDP cluster.
Support Mixed-case Identifiers	Select **Yes** from the drop down list.

Property	Description
Use Sqoop Connector	Select **Sqoop v1.x** to enable Sqoop connectivity.

Sqoop Arguments	Sqoop argument required to connect to the database.
	Use the following argument string format to connect from a Cloudera CDH cluster:
	`-Dsqoop.connection.factories=com.cloudera.connector.teradata.TeradataManagerFactory`
	Use the following argument string format to connect from a Hortonworks HDP cluster:
	`-Dsqoop.connection.factories=org.apache.sqoop.teradata.TeradataManagerFactory`

6. When you finish entering the connection properties, click **Test Connection** to test the connection.
7. Click **Finish**.

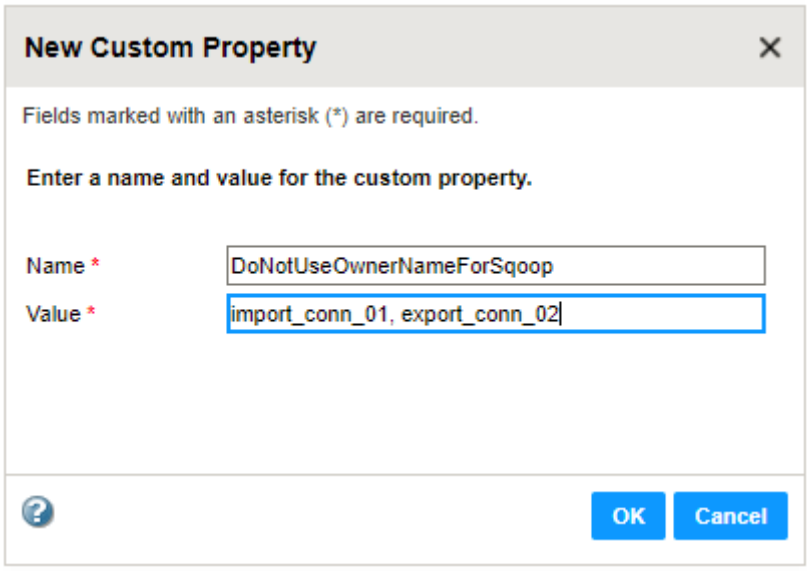
Configure the DoNotUseOwnerNameForSqoop Custom Option

Use the Administrator tool to add the DoNotUseOwnerNameForSqoop custom property to the Enterprise Data Preparation Service. Enter the name of each JDBC connection that Enterprise Data Preparation uses to import and export data as the custom property value.

The DoNotUseOwnerNameForSqoop custom property enables Sqoop to use the connections to access a table in the Teradata database without requiring the name of the schema that contains the table. By default, Sqoop requires the name of the schema that contains the table.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the Enterprise Data Preparation Service.
3. Click the **Properties** view.
4. Click the pencil icon in the Custom Options section.
5. Click **New** in the Edit Custom Options dialog box.
6. Enter DoNotUseOwnerNameForSqoop as the custom property name.
7. Enter the name of each JDBC connection Enterprise Data Preparation uses to import and export data as the custom property value. If you enter multiple connection names, separate each connection name with a comma.

The following image shows the **New Custom Property** dialog box:



8. Click **OK**.

You can now import and export data assets between Enterprise Data Preparation and Teradata.

For more information about importing and exporting data assets, see the "Import, Export, and Upload Data" chapter in the Enterprise Data Preparation User Guide.

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